Knowledge Graphs for Healthcare

Ontology + data + reasoning — and what it means for care

M Usman · Semantic Web · MSc Computing, University of Huddersfield

Contents

- **About KG** — introduction to knowledge graphs and ontologies
- **About Healthcare** — knowledge graphs in healthcare
- **Applications** — general and specific-purpose ontologies
- **Benefits and way forward** — the healthcare future

1 · About KG

Introduction to knowledge graphs and ontologies

M Usman · Knowledge Graphs for Healthcare

Why KG matters in healthcare

Healthcare turns on three functions:

- **Proper diagnosis**
- **Prevention**
- **Treatment**

Each is constrained by the same three things: **time**, **accuracy**, and **coverage**.

Typical healthcare functions.

Why KG matters in healthcare

Powered by a knowledge graph, the same functions — diagnosis, **prediction**, treatment — improve on exactly those constraints:

- **Time** — faster retrieval and reasoning
- **Accuracy** — connected, validated knowledge
- **Coverage** — breadth across sources

Toward a pervasive healthcare system.

What is a knowledge graph?

> A knowledge graph acquires and integrates information into an **ontology** and applies a **reasoner** to derive new knowledge.

In short:

Knowledge representation (KR) + reasoning (R) → knowledge graph

KG and ontology

Knowledge graph

- Datasets drawn from multiple sources
- Uncovers connections not seen before

Ontology

- An organised knowledge base
- A formal representation of entities as a graph

ontology + data = knowledge graph

Ontology and its origins

Aristotle — 384–322 BC

- First systematic taxonomy of biology
- Classification of organisms by shared properties

Galen — 130–210 AD

- Systematic description of diseases, signs, and symptoms

KG architecture

A knowledge-base system:

- **Sources** (Source-1, Source-2, Source-3, ...)
- feed a **knowledge base** (for example, an ontology)
- which a **reasoning engine** queries
- to produce the **knowledge graph**

2 · About Healthcare

Healthcare graphs and the UK context

M Usman · Knowledge Graphs for Healthcare

The UK healthcare system

- The **NHS** was established in 1948; comparable systems exist worldwide.
- Predictive, preventive, and personalised medicine is enabled through telemedicine.
- Prevention is strong in the UK; prediction and personalisation are growing.
- Scientific discovery and public involvement both drive predictive and personalised care.

Tiers: primary care (community, GPs, dentists, pharmacists) · secondary care (hospital, via GP referral) · tertiary care (specialist hospitals).

Ontologies in biomedicine

Three pillars:

- **Gene ontologies**
- **Disease ontologies**
- **Phenotype ontologies**

As of September 2021: **BioPortal** held 934 ontologies; the **OBO Foundry** held 234.

Seven steps to a healthcare ontology

1. **Scope** — identify the domain and scope
2. **Re-use** — consider existing ontologies
3. **Critical terms** — enumerate the important terms
4. **Classes and hierarchy**
5. **Properties of classes**
6. **Define sub-classes**
7. **Create instances**

Ontology deployment architecture

Biomedical ontologies → **model mapping** → **ontologies** → **applications** (external and internal).

Reference sources:

- OBO Foundry — `obofoundry.org`
- BioPortal — `bioportal.bioontology.org`

3 · Applications

General and specific-purpose ontologies

M Usman · Knowledge Graphs for Healthcare

Gene Ontology (GO)

GO defines gene function across three types:

- **Biological process** — pathways and processes genes act on
- **Cellular component** — the parts a gene relates to
- **Molecular function** — the composition of genes

GO promotes interoperability of gene function across species.

3,654,519

total gene products catalogued in GO

SNOMED CT

- A controlled, coded clinical terminology for electronic health records.
- Built by the UK NHS and the College of American Pathologists (US, 2002).
- Maintained by the IHTSDO, which adopted the trading name **SNOMED International** in 2017.
- Supports information retrieval, decision support, communication, and pattern identification.

Systematized Nomenclature of Medicine — Clinical Terms. 19 top classes; sub-classes exceed 361,569.

SNOMED CT

- One logical framework for clinical concepts: diagnoses, signs, and symptoms.
- Covers procedures — surgical, therapeutic, diagnostic — and observables such as heart rate.
- Reduces the need for multiple, overlapping code systems.

40 countries are members of SNOMED International.

General-purpose ontologies

[Visual summary in the original deck — supply the general-purpose-ontologies graphic here.]

GO and SNOMED CT are broad, reusable foundations. The next examples are purpose-built.

Specific purpose: the COVID-19 Knowledge Graph (CKG)

A document-similarity engine that combines **semantic** and **relationship** information from related COVID-19 research.

CKG visualisation: paper entities connect to concepts, topics, and authors through directed relations; authors connect to institutions.

CKG web search interface

[Screenshot in the original deck — supply the CKG web-search interface image here.]

A research-facing search built directly on the knowledge graph.

DRANPTO — a dementia-agitation ontology

A specialised ontology for non-pharmacological treatment of agitation in dementia.

DRANPTO has 11 top classes and 558 sub-classes.

Challenges — DRANPTO

- **Knowledge base** — no standard base; a PRISMA-based approach gathers the information
- **Manual process** — automation is not adopted; non-taxonomic relations are missed by automatic methods
- **Ontology evaluation** — gold-standard, domain-expert, and competency-question methods
- **Expert evaluation** — qualified professionals are scarce, and the work is time-consuming

Automated vs manual ontology creation

CKG is built automatically by machine learning for **research**; DRANPTO is built manually for a **clinical** setting.

Automated — misses non-taxonomic relations · faster to build · maintained by ML
Manual — can express non-taxonomic relations · slower · limited by expert availability

Proposal: a semi-automatic process that takes the best of both.

Challenges — matching ontologies

Ontology matching and fusion combines standards into one graph:

- **SNOMED CT** (clinical terms) · **DXplain** (diagnostic concepts)
- **LOINC** (lab) · **RxNorm** (drugs and chemicals) · Gene Ontology and others

With **human feedback**, these fuse into a single knowledge graph.

Challenges — ontology versions

A historical knowledge graph improves over time:

- Better information retrieval
- Maintained semantic annotations as the ontology evolves

4 · Benefits and way forward

The healthcare future

Benefits of KG in healthcare

- **Doctors** — knowledge extraction, reliable information, decision-making
- **Patients** — better outcomes, lower cost
- **Service providers** — predictive, preventive, personalised services
- **Industry** — discoveries, innovation, research
- **Public health policy** — data-based policy

Way forward

Major challenges

- Ontologies need constant upgrading as knowledge advances
- Space and time are not yet well mapped across the knowledge base

The future

- Prediction and personalisation of care
- Pervasive, ontology-enabled healthcare
- **Healthcare 4.0**, enabled by IoT

Conclusion

- **Creating KGs** — acquire and integrate knowledge; organise it through an ontology
- **Maintaining KGs** — store and access; the physical side of keeping a graph current
- **Exploiting KGs** — understand and consume knowledge through better search, summarisation, and question answering

References

Alonso (2020) · Cardoso, Silveira & Pruski (2020) · Dahbi, Lamharhar & Chiadmi (2020) · Ehrlinger & Wöß (2016) · Grosios, Gahan & Burbidge (2010) · Noy & McGuinness (2021) · Pan et al. (2017) · Schulz, Stenzhorn, Boeker & Smith (2009) · Tiwari, Al-Aswadi & Gaurav (2021) · Wise et al. (2020) · Yang et al. (2021) · Zeshan & Mohamed (2012) · Zhang et al. (2020).

Full citations in the speaker notes / paper.

Thank you

M Usman · Data Architect & Applied-AI Researcher

About KG · About Healthcare · Applications · Benefits and way forward

See the full portfolio

The work behind this deck sits on one page: the Insights series, the projects and the code that runs them, and the publications.

mtusman-ai.github.io/M.Usman

M Usman · From enterprise data to production AI.